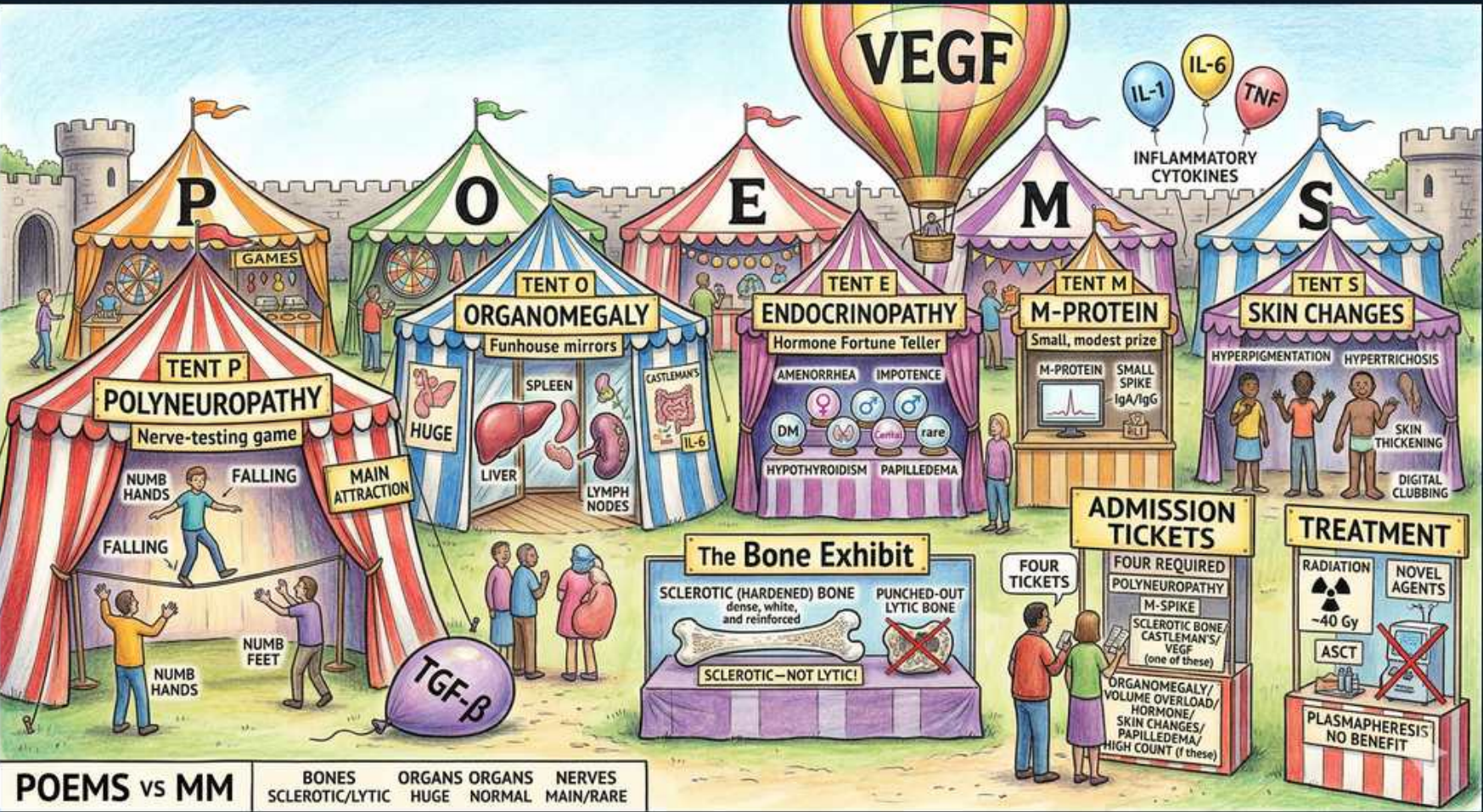


POEMS Syndrome — The Carnival Tents



1. P — Polyneuropathy

WHAT YOU SEE

Tent P: numb hands and feet, figure falling.

WHAT IT ENCODES

Mandatory criterion. A chronic, progressive, demyelinating sensorimotor polyneuropathy (CIDP-like) is the dominant and usually presenting feature of POEMS.

BOARD TRAP

Polyneuropathy is REQUIRED — a POEMS diagnosis without neuropathy is wrong. It is symmetric and demyelinating, mimicking CIDP.

2. O — Organomegaly

WHAT YOU SEE

Tent O: huge liver, spleen, lymph nodes.

WHAT IT ENCODES

Hepatomegaly, splenomegaly, and/or lymphadenopathy. Lymph node biopsy may show Castleman disease, which is associated with POEMS.

BOARD TRAP

Castleman disease can coexist with POEMS — its presence is a major associated criterion, not a separate unrelated finding.

3. E — Endocrinopathy

WHAT YOU SEE

Tent E: fortune teller, hormone signs.

WHAT IT ENCODES

Multiple endocrinopathies: hypogonadism (impotence, gynecomastia), hypothyroidism, adrenal insufficiency, diabetes. At least one (other than diabetes/hypothyroid alone) is needed.

BOARD TRAP

Diabetes or hypothyroidism alone is too common to count — POEMS needs additional endocrine axes involved.

4. M — Monoclonal plasma cell disorder

WHAT YOU SEE

Tent M: small modest M-protein prize.

WHAT IT ENCODES

Mandatory criterion. A clonal plasma-cell disorder, almost always LAMBDA light-chain restricted, usually with a SMALL M-protein. Often a sclerotic plasmacytoma.

BOARD TRAP

M-protein in POEMS is small and lambda-restricted — a big spike suggests myeloma instead. The clone is the driver, not the bulk.

5. S — Skin changes

WHAT YOU SEE

Tent S: hyperpigmentation, thick skin, clubbing.

WHAT IT ENCODES

Hyperpigmentation, hypertrichosis, skin thickening, digital clubbing, glomeruloid hemangiomas. Reflects the cytokine/VEGF-driven systemic process.

BOARD TRAP

Skin changes are a minor criterion; they support but do not by themselves establish POEMS.

6. VEGF elevation

WHAT YOU SEE

VEGF hot-air balloon rising over the carnival.

WHAT IT ENCODES

Markedly elevated VEGF is the key mediator of POEMS, driving edema, organomegaly, skin changes, and capillary leak. Serum VEGF is a major diagnostic criterion and tracks disease activity.

BOARD TRAP

VEGF — not IL-6 — is the central POEMS biomarker and treatment-response marker; high VEGF is a major criterion.

7. Sclerotic bone lesions

WHAT YOU SEE

Bone exhibit: sclerotic ('hardened') bone.

WHAT IT ENCODES

POEMS bone lesions are OSTEOSCLEROTIC (dense, reinforced), unlike myeloma's punched-out lytic lesions. A sclerotic plasmacytoma is a common source of the clone.

BOARD TRAP

Sclerotic (not lytic) bone lesions distinguish POEMS from multiple myeloma — this is a classic discriminator.

8. Inflammatory cytokines (IL-1, IL-6, TNF)

WHAT YOU SEE

Cloud of IL-1, IL-6, TNF cytokines.

WHAT IT ENCODES

Proinflammatory cytokines accompany VEGF and contribute to systemic features and Castleman overlap, but VEGF dominates the syndrome's pathophysiology.

BOARD TRAP

IL-6 drives Castleman; VEGF drives POEMS — don't swap the headline cytokine when asked the key mediator.

9. Diagnostic criteria (need both mandatory + others)

WHAT YOU SEE

Admission tickets: four required.

WHAT IT ENCODES

Diagnosis = both mandatory criteria (polyneuropathy + monoclonal plasma-cell disorder) plus at least one major (sclerotic bone, Castleman, or high VEGF) plus minor criteria.

BOARD TRAP

You cannot diagnose on the acronym alone — both mandatory criteria plus a major criterion are required.

10. Extravascular volume overload

WHAT YOU SEE

Funhouse mirrors distorting figures (edema).

WHAT IT ENCODES

VEGF-driven capillary leak causes peripheral edema, ascites, and pleural effusions — an additional supporting feature reflecting vascular permeability.

BOARD TRAP

The edema/effusions are VEGF-mediated capillary leak, not heart failure or nephrotic syndrome.

11. Thrombocytosis / erythrocytosis

WHAT YOU SEE

Crowded blood-count booth.

WHAT IT ENCODES

POEMS classically causes HIGH platelets and HIGH red-cell counts — opposite of the cytopenias seen in myeloma. A useful supporting clue.

BOARD TRAP

Thrombocytosis/erythrocytosis (not cytopenia) favors POEMS over myeloma — reversed marrow picture.

12. Radiation for solitary sclerotic lesion

WHAT YOU SEE

Treatment booth: radiation, ~40 Gy.

WHAT IT ENCODES

When there is a single/limited osteosclerotic plasmacytoma, localized radiotherapy can produce durable responses and neuropathy improvement.

BOARD TRAP

For solitary sclerotic lesion → radiation, not systemic chemo; reserve systemic therapy for disseminated marrow disease.

13. Systemic therapy / ASCT

WHAT YOU SEE

ASCT and novel-agent options.

WHAT IT ENCODES

For disseminated POEMS (diffuse marrow plasma cells), treat the clone with melphalan-based therapy and autologous stem-cell transplant, often improving VEGF and neuropathy.

BOARD TRAP

Disseminated marrow disease needs systemic anti-plasma-cell therapy/ASCT — radiation alone won't control it.

14. Plasmapheresis NOT beneficial

WHAT YOU SEE

Note: plasmapheresis — no benefit.

WHAT IT ENCODES

Unlike many paraprotein neuropathies, plasmapheresis does NOT help POEMS because the M-protein is small and VEGF/cytokines drive disease. Treat the clone instead.

BOARD TRAP

Plasmapheresis is useless in POEMS — a deliberate trap given how often it's used in other paraprotein disorders.

15. POEMS vs MM discriminators

WHAT YOU SEE

Comparison banner: bones, organs, nerves.

WHAT IT ENCODES

POEMS: sclerotic bone, organomegaly, prominent neuropathy, small lambda M-protein, high VEGF/platelets. MM: lytic bone, hypercalcemia, renal failure, large M-protein, cytopenias.

BOARD TRAP

The fastest discriminator: sclerotic+neuropathy+lambda+thrombocytosis = POEMS; lytic+CRAB = myeloma.